

Dominika Radziun

Education and Professional Experience

- 2023–present **Postdoctoral Researcher**
Donders Institute for Brain, Cognition, and Behaviour, Nijmegen, the Netherlands
Advisor: Dr. Luke E. Miller
- May 2026 **Guest Researcher**
IMT School for Advanced Studies Lucca, Italy
- 2018–2023 **PhD in Neuroscience**
Karolinska Institutet, Stockholm, Sweden
Thesis title: *Body perception and brain plasticity in blind and sighted individuals*
Supervisors: Prof. Henrik Ehrsson (KI), Prof. Marcin Szwed (Jagiellonian University),
Prof. Amir Amedi (Reichman University), Dr. Laura Crucianelli (KI)
External examiner: Prof. Brigitte Röder (University of Hamburg)
- 2013–2018 **MA in Psychology**
Jagiellonian University, Kraków, Poland
Inter-faculty Individual Studies in the Humanities and Social Sciences
Supervisors: Prof. Henrik Ehrsson (Karolinska Institutet) and Prof. Marcin Szwed (JU)
- Nov 2016–
Jan 2017 **Visiting Student**
University of Cambridge, United Kingdom
Department of Psychology, Behavioural and Clinical Neuroscience Institute
- Jan 2016–
May 2016 **Visiting Student**
University of Oslo, Norway
Department of Psychology
Department of Philosophy, History of Art and Ideas, Greek and Latin

Journal Articles and Preprints

- 2026 **Radziun, D.**, Schippers, A., Longo, M. R., & Miller, L. E. (2026). Sensorimotor training lightens the perceived weight of body augmentation devices. *bioRxiv*. [\[Link\]](#)
- Radziun, D.**, & Larsson, D. E. (2026). Breath, Belief, and Brain: A Tripartite Map of Respiratory Interoception. *Journal of Neuroscience*, 46(13), e1891252026. [\[Link\]](#)
- Radziun, D.***, Geurts, S.*, Peviani, V. C., & Miller, L. E. (2026). Dynamics of sensorimotor plasticity during spatial finger augmentation. *eLife*, 15. [\[Link\]](#)
- 2025 **Radziun, D.**, & Miller, L. E. (2025). Vision and body representation: Insights from blindness. *PsyArXiv*. [\[Link\]](#)
- 2024 Stroh, A. L.*, **Radziun, D.***, Korczyk, M., Crucianelli, L., Ehrsson, H. H., & Szwed, M. (2024). Blind individuals' enhanced ability to sense their own heartbeat is related to the thickness of their occipital cortex. *Cerebral Cortex*, 34(8). [\[Link\]](#)
- Radziun, D.**, Korczyk, M., Szwed, M., & Ehrsson, H. H. (2024). Are blind individuals immune to bodily illusions? Somatic rubber hand illusion in the blind revisited. *Behavioural Brain Research*, 460, 114818. [\[Link\]](#)
- Crucianelli, L., **Radziun, D.**, & Ehrsson, H. H. (2024). Thermosensation and emotion: Thermosensory accuracy in a dynamic thermal matching task is linked to depression and anxiety symptomatology. *Physiology & Behavior*, 273, 114407. [\[Link\]](#)
- 2023 **Radziun, D.**, Korczyk, M., Crucianelli, L., Szwed, M., & Ehrsson, H. H. (2023). Heartbeat counting accuracy is enhanced in blind individuals. *Journal of Experimental Psychology: General*, 152(7), 2026–2039. [\[Link\]](#)
- Radziun, D.**, Crucianelli, L., Korczyk, M., Szwed, M., & Ehrsson, H. H. (2023). The perception of affective and discriminative touch in blind individuals. *Behavioural Brain Research*, 444, 114361. [\[Link\]](#)

- 2022 **Radziun, D.**, Crucianelli, L., & Ehrsson, H. H. (2022). Limits of cross-modal plasticity? Short-term visual deprivation does not enhance cardiac interoception, thermosensation, or tactile spatial acuity. *Biological Psychology*, 168, 108248. [\[Link\]](#)
- Ehrsson, H. H., Fotopoulou, A., **Radziun, D.**, Longo, M. R., & Tsakiris, M. (2022). No specific relationship between hypnotic suggestibility and the rubber hand illusion. *Nature Communications*, 13(1). [\[Link\]](#)
- 2018 **Radziun, D.**, & Ehrsson, H. H. (2018). Auditory cues influence the rubber-hand illusion. *Journal of Experimental Psychology: Human Perception and Performance*, 44(7), 1012–1021. [\[Link\]](#)
- Radziun, D.**, & Ehrsson, H. H. (2018). Short-term visual deprivation boosts the flexibility of body representation. *Scientific Reports*, 8(1). [\[Link\]](#)
- 2017 Westerhausen, R., Pöldver, N., Naar, R., **Radziun, D.**, Kaarep, M. S., Kreegipuu, K., ... Kompus, K. (2017). Effect of voicing on perceptual auditory laterality in Estonian and Norwegian native speakers. *Applied Psycholinguistics*, 39(2), 259–273. [\[Link\]](#)
- Bola, Ł., **Radziun, D.**, Siuda-Krzywicka, K., Sowa, J. E., Paplińska, M., Sumera, E., & Szwed, M. (2017). Universal Visual Features Might Be Necessary for Fluent Reading. A Longitudinal Study of Visual Reading in Braille and Cyrillic Alphabets. *Frontiers in Psychology*, 8. [\[Link\]](#)

Awards and Scholarships

- 2026 Society for the Neural Control of Movement Conference Scholarship (\$800)
- 2024 Glushko Dissertation Prize (\$10,000)
Awarded by the Cognitive Science Society and the Glushko-Samuelson Foundation for an outstanding PhD thesis in cognitive science
- 2023 Poster Prize, Research in Touch & International Association for the Study of Affective Touch
- 2017, 2018 Polish Ministry of Higher Education Scholarship for outstanding students (€3600 annually)
- 2017, 2018 Jagiellonian University Chancellor Scholarship for the best students (€1500 annually)
- 2013–2018 Pro Akademia Foundation Scholarship for academic excellence and social involvement (€1100 annually)
- 2013–2014 Educational Enterprise Foundation Scholarship (€1200 annually)

Invited and Prize Talks

- May 2026 *Building and rebuilding models of self and world: From sensory plasticity to psychosis resilience*. Invited talk, IMT Lucca, May 6th, Lucca, Italy
- July 2024 *Body perception and brain plasticity in blind and sighted individuals*. Glushko Dissertation Prize talk, the 46th Annual Meeting of the Cognitive Science Society, July 27th 2024, Rotterdam, the Netherlands
- April 2024 *Body perception and brain plasticity in blind and sighted individuals*. Invited talk, Department of Psychology, University of Milano-Bicocca, April 29th 2024, Milan, Italy
- Sept 2022 *The perception of interoceptive signals in blind individuals*. Invited talk, Disability Research Division, Linköping University, September 14th 2022, Linköping, Sweden
- June 2022 *The perception of interoceptive signals in blind individuals*. Invited talk, Clinical and Affective Neuroscience Group, University College London, June 9th 2022, London, United Kingdom (remote)

Conference Talks

- June 2026 *Cardiac interoception and occipital cortex plasticity in blindness*. 8th European Society for Cognitive and Affective Neuroscience Meeting, June 5th 2026, Rome, Italy
- April 2026 *The influence of using finger-extending exoskeletons on proprioceptive and tactile localization*. Neural Control of Movement Conference, April 24th 2026, Kobe, Japan

- July 2025 *Cross-modal plasticity in the occipital cortex supports interoceptive processing in blindness.* Symposium talk, 23rd International Multisensory Research Forum, June 16th 2025, Durham, United Kingdom
- Sept 2023 *Body perception and brain plasticity in blind and sighted individuals.* 5th Body Representation Network Conference, September 15th 2023, Palma de Mallorca, Spain
- Sept 2023 *Enhanced cardiac interoception in blind individuals: unraveling cross-modal plasticity and emotional processing.* Symposium talk, 23rd Conference of the European Society for Cognitive Psychology, September 9th 2023, Porto, Portugal
- Sept 2023 *Body perception and brain plasticity in blind and sighted individuals: from heartbeats to rubber hands.* 23rd Conference of the European Society for Cognitive Psychology, September 7th 2023, Porto, Portugal
- June 2023 *Body perception and brain plasticity in blind and sighted individuals: from heartbeats to rubber hands.* 21st International Multisensory Research Forum, June 30th 2023, Brussels, Belgium
- Oct 2022 *The perception of interoceptive signals in blind individuals.* 12th Neuronus IBRO Neuroscience Forum, October 17th 2022, Kraków, Poland
- July 2022 *The perception of interoceptive signals in blind individuals.* 6th European Society for Cognitive and Affective Neuroscience Meeting, July 19th 2022, Vienna, Austria
- March 2022 *Cardiac interoception and affective touch sensitivity are enhanced in the blind.* 9th Mind, Brain and Body Symposium, March 16th 2022, online
- June 2021 *Physiologically implausible warmth, but not cold, suppresses the feeling of limb ownership in the somatic rubber hand illusion.* Short talk given at the 24th Annual Meeting of the Association for the Scientific Study of Consciousness, June 15th 2021, online
- March 2016 *Diagnoza inteligencji u osób niewidomych. Możliwości i ograniczenia* [Intelligence testing in blind individuals. Possibilities and limitations]. 8th Cracow Cognitive Science Conference, March 19–20th 2016, Kraków, Poland

Poster Presentations

- July 2025 *Neurophysiological effects of training with a finger-extending exoskeleton: An EEG study.* 7th Body Representation Network Conference, July 4th 2025, Utrecht, the Netherlands
- March 2025 *The influence of using finger-extending exoskeletons on tactile and proprioceptive localization.* 12th Mind, Brain and Body Symposium, March 11th 2025, Berlin, Germany
- Sept 2024 *The influence of using finger-extending exoskeletons on tactile localization.* 6th Body Representation Network Conference, September 13th 2024, Verona, Italy
- July 2023 *The perception of affective and discriminative touch in blind individuals.* Festival of Touch Conference, July 4–7th 2023, Marseille, France | **Poster Prize**
- July 2022 *The perception of interoceptive signals in blind individuals.* 25th Annual Meeting of the Association for the Scientific Study of Consciousness, July 12–15th 2022, Amsterdam, the Netherlands
- March 2022 *Cardiac interoception and affective touch sensitivity are enhanced in the blind. Emotional and mental health implications of increased sensory acuity.* Society for Affective Science Interoception Preconference, March 30th 2022, online
- Feb 2022 *Cardiac interoception and affective touch sensitivity are enhanced in blind individuals.* The Brain Conference 2022, February 24–25th 2022, online
- Oct 2018 *Short-term visual deprivation boosts the flexibility of body representation.* Blind Brain Workshop, October 11–13th 2018, Lucca, Italy
- June 2018 *Short-term visual deprivation boosts the flexibility of body representation.* 22nd Annual Meeting of the Association for the Scientific Study of Consciousness, June 26–29th 2018, Kraków, Poland
- April 2018 *The influence of short-term visual deprivation on body representation.* Neuronus IBRO & IRUN Neuroscience Forum, April 20–22nd 2018, Kraków, Poland
- June 2017 *Auditory Cues Influence the Rubber-Hand Illusion.* 2nd Nordic Neuroscience Meeting, June 7–9th 2017, Stockholm, Sweden

- March 2017 *The role of visual deprivation in bodily awareness.* 5th Mind, Brain and Body Symposium, March 16–17th 2017, Berlin, Germany
- April 2016 *Are universal visual features necessary for reading? Preliminary results of a longitudinal study of visual Braille and Cyrillic alphabets' learners.* Neuronus IBRO & IRUN Neuroscience Forum, April 22–24th 2016, Kraków, Poland
- April 2015 *Age-related alterations in attentional set-shifting.* Neuronus IBRO & IRUN Neuroscience Forum, April 17–19th 2015, Kraków, Poland

Pre-doctoral Research and Clinical Experience

- 2017–2018 **Clinical Psychology Intern, Dr. Józef Babiński Clinical Hospital, Kraków, Poland**
Assisted in psychological assessments and patient observations under supervision.
Took part in therapeutic sessions and group activities.
Gained experience in case documentation and interdisciplinary team collaboration.
Observed the application of psychiatric treatments, including pharmacotherapy and psychotherapy.
- 2015–2018 **Research Intern, Jagiellonian University, Kraków, Poland**
Supervisor: Prof. Marcin Szwed
Project: Are universal visual features necessary for fluent reading? Longitudinal study of visual reading in Braille and Cyrillic alphabets
- 2017 **Research Intern (thesis internship), Karolinska Institutet, Stockholm, Sweden**
Supervisor: Prof. Henrik Ehrsson
Project: The influence of short-term visual deprivation on body representation
- 2016–2017 **Research Intern, University of Cambridge, United Kingdom**
Supervisors: Dr. Maria Niedernhuber, Prof. Tristan Bekinschtein
Project: Investigating conscious processes underlying the prediction of sensory inputs in the somatosensory and auditory domains – an EEG study
- 2016 **Research Intern, Karolinska Institutet, Stockholm, Sweden**
Supervisors: Dr. Björn van der Hoort, Prof. Henrik Ehrsson
Project: Body ownership and visual awareness – a continuous flash suppression study
- 2016 **Research Intern, University of Oslo, Norway**
Supervisor: Prof. René Westerhausen
Project: Effect of voicing on perceptual auditory laterality in Estonian and Norwegian native speakers

Supervision

- May 2025–
present **Eleni Manias**
Master's in Cognitive Neuroscience, Radboud University (thesis)
- Sept 2024–
March 2025 **David Teulings**
Master's in Artificial Intelligence, Radboud University (thesis)
- Sept 2024–
March 2025 **Arina Schippers**
Master's in Artificial Intelligence, Radboud University (internship)
- March 2024–
Dec 2024 **Siebe Geurts**
Master's in Artificial Intelligence, Radboud University (thesis)

Teaching

- May 2025–
May 2026 **Guest Lecturer**
Sensorimotor Neurotechnology course (lectures on Body Augmentation and Embodiment) for Artificial Intelligence and Cognitive Neuroscience master's students at Radboud University
- Oct 2020–
March 2022 **Teaching Assistant**
Neuroanatomy workshop for medical students at Karolinska Institutet

Reviewing

Journal *Biological Psychology; Cognition; Consciousness & Cognition; Cortex; Disability and Rehabilitation: Assistive Technology; Frontiers in Psychology; Heliyon; Human Brain Mapping; Journal of Anxiety Disorders; Journal of Behavior Therapy and Experimental Psychiatry; Journal of Experimental Psychology: Human Perception and Performance; Journal of Psychiatric Research; Nature Human Behaviour; Perception; Psychophysiology; PLOS ONE; Scientific Reports; Translational Psychiatry; Transactions on Haptics; Virtual Reality*

Grant National Science Centre (Poland)

Community Service

Sept 2025– **Member, Radboud Young Academy**

present Selected member of an interdisciplinary think tank of early- and mid-career staff across Radboud University and Radboudumc, contributing to discussions on academic culture, leadership, and research policy.

Sept 2024– **Postdoc Representative, Donders Postdoc Council**

present Organize events, relay concerns to institute leadership, and support initiatives related to professional development and working conditions.

Jan 2024– **Co-Chair, Halkes Women+ Faculty Network**

June 2026 Coordinate activities supporting women and gender minority faculty and represent the network in meetings with university leadership.

April 2024– **Member, Donders Sessions and Lectures Committee**

January 2025 Co-organized the event *Bridging Science and Climate Activism* (with the CNS Green Team).

October 2023– **Conference Co-Organizer, Body Representation Network**

July 2025 Coordinated speaker communication and supported planning and logistics.

Dec 2017– **Communications Officer & Volunteer Coordinator, Association for the Scientific Study of Consciousness**

Dec 2023 Managed social media and newsletter; coordinated volunteers for ASSC 22 (2018, Kraków).

March–July **Co-Coordinator (Project Fleck), Collegium Invisibile**

2022 Co-organized a mentoring initiative supporting Ukrainian students displaced by war in accessing and continuing higher education in Poland and abroad.

Oct 2015– **Tutor & Project Coordinator (First Tutor), Collegium Invisibile & Polish Association for Gifted Youth**

Sept 2022

Taught introductory neuroscience and supervised high-school students (1–3 annually). Coordinated the First Tutor program (2016–2021) and co-organized two academic summer schools (2018, 2019), where I also taught a course on consciousness research.

Media Coverage

2023 U.S. News & World Report, *Blind People Are Better at Sensing Their Heartbeats*

ScienceAlert, *Blind People Possess an Acute Awareness of Their Own Heartbeat*

MedicalXpress, *Blind people sense their heartbeats better than sighted people, shows study*

Sveriges Radio, *Blinda känner sina hjärtslag bättre*

Deutschlandfunk Nova, *Studie: Blinde Menschen spüren ihren Herzschlag besser als Sehende*

Ça m'intéresse, *Les malvoyants ont cette incroyable capacité que les voyants n'ont pas*

CNN Indonesia, *Pakar Ungkap 'Kekuatan Super' Penyandang Tuna Netra*